Types in Multivariable Calculus

Reading the Notation of Line, Surface, and Vector Integrals

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Abstract

The notation of vector calculus is dense because it hides the *types* of its objects. A single symbol like $\mathrm{d}\mathbf{S}$ or $\nabla \cdot \mathbf{F}$ silently declares whether something is a number or a vector, and whether an operation consumes a field and returns a field or a scalar. This note treats every object as if it had a function signature: domain in, codomain out. We unpack each differential element ($\mathrm{d}s$, $\mathrm{d}\mathbf{r}$, $\mathrm{d}S$, $\mathrm{d}\mathbf{S}$, $\mathrm{d}V$), each operator (∇, $\nabla\cdot$, $\nabla\times$), and each integral, and we show that every one of them collapses, after a parametrization, into an ordinary single-variable Riemann integral. The recurring theme: *a parametrization is a type cast* that turns a geometric integral into the $\int_a^b (\cdots)\,\mathrm{d}t$ you already know.

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1 The typing problem

A function signature tells you, before you read a single line of the body, what goes in and what comes out: `length : string -> int`. You know immediately that feeding it an `int` is an error, and that the result is a number you can add to other numbers.

Vector-calculus notation has the opposite ergonomics. The expression

$$\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S}$$

declares its types only implicitly. To read it you must already know that S is a *surface*, that $\mathbf{F}$ is a *vector field*, that $\mathrm{d}\mathbf{S}$ is an *infinitesimal vector* (not a scalar), that "$\cdot$" is the dot product (which forces the result of the integrand to be a *scalar*), and therefore that the whole expression evaluates to a single *number*. None of that is written down; you reconstruct it in your head.

This document makes the reconstruction explicit. For every piece of notation we state, in the style of a function signature, what type each symbol carries and what type the combination produces.

1.1 The type vocabulary

We will work in $\mathbb{R}^2$ or $\mathbb{R}^3$. The objects fall into a small number of types.

Type	**Notation**	**What it is**
point	$\mathbf{p} = (x, y, z) \in \mathbb{R}^3$	a location in space
scalar	$c \in \mathbb{R}$	a single number
vector	$\mathbf{v} \in \mathbb{R}^3$	a magnitude-and-direction arrow
scalar field	$f : \mathbb{R}^3 \to \mathbb{R}$	a number at every point
vector field	$\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$	an arrow at every point
parametrized curve	$\mathbf{r} : [a, b] \to \mathbb{R}^3$	a 1-parameter path
parametrized surface	$\mathbf{r} : D \subset \mathbb{R}^2 \to \mathbb{R}^3$	a 2-parameter sheet

Two distinctions do almost all the work:

- **scalar vs. vector.** A scalar field assigns a *number* (temperature, pressure, potential). A vector field assigns an *arrow* (velocity, force, flow). Confusing the two is the single most common reading error.

- **point vs. vector.** A point is *where* you are; a vector is a *displacement* or arrow. They live in the same $\mathbb{R}^3$ and are written the same way, which is exactly why the notation is slippery. A field eats a point and returns either a scalar or a vector.

2 Differential elements, built from single-variable calculus

Every integral below is glued together from an *infinitesimal element*: $\mathrm{d}s$, $\mathrm{d}\mathbf{r}$, $\mathrm{d}S$, $\mathrm{d}\mathbf{S}$, or $\mathrm{d}V$. Each one has a type, and each one is derived by the same move you already use in one dimension, namely $\mathrm{d}u = u'(t)\,\mathrm{d}t$.

2.1 The scalar arc-length element $\mathrm{d}s$

Type: scalar. $\mathrm{d}s$ is an infinitesimal *length*, a non-negative number.

Start from the Pythagorean theorem applied to an infinitesimal right triangle whose legs are $\mathrm{d}x$ and $\mathrm{d}y$:

$$\mathrm{d}s = \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} \qquad \big(\text{in } \mathbb{R}^3:\ \mathrm{d}s = \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2}\big).$$

This is just $a^2 + b^2 = c^2$ for the hypotenuse of a tiny triangle. To make it computable, introduce a parameter t with $\mathbf{r}(t) = (x(t), y(t))$. Then each differential is an ordinary single-variable differential,

$$\mathrm{d}x = x'(t)\,\mathrm{d}t, \qquad \mathrm{d}y = y'(t)\,\mathrm{d}t,$$

and substituting,

$$\mathrm{d}s = \sqrt{x'(t)^2\,\mathrm{d}t^2 + y'(t)^2\,\mathrm{d}t^2} = \sqrt{x'(t)^2 + y'(t)^2}\;\mathrm{d}t = |\mathbf{r}'(t)|\,\mathrm{d}t.$$

The factor $|\mathbf{r}'(t)|$ is the *speed*: a scalar. This is the whole trick in miniature. The geometric object $\mathrm{d}s$ becomes "speed times $\mathrm{d}t$", which is exactly the kind of thing you integrate in first-year calculus.

2.2 The vector line element $\mathrm{d}\mathbf{r}$

Type: vector. $\mathrm{d}\mathbf{r}$ is an infinitesimal *displacement arrow*, pointing along the curve in the direction of increasing parameter.

Where $\mathrm{d}s$ keeps only the length, $\mathrm{d}\mathbf{r}$ keeps the components:

$$\mathrm{d}\mathbf{r} = (\mathrm{d}x,\ \mathrm{d}y,\ \mathrm{d}z) = \big(x'(t),\ y'(t),\ z'(t)\big)\,\mathrm{d}t = \mathbf{r}'(t)\,\mathrm{d}t.$$

So $\mathrm{d}\mathbf{r}$ is the velocity vector $\mathbf{r}'(t)$ scaled by $\mathrm{d}t$. Its length is $|\mathrm{d}\mathbf{r}| = |\mathbf{r}'(t)|\,\mathrm{d}t = \mathrm{d}s$, which gives the bridge between the two:

$$\boxed{\mathrm{d}\mathbf{r} = \mathbf{T}\,\mathrm{d}s} \qquad \text{where} \qquad \mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \ \text{ is the unit tangent (a vector).}$$

Read the boxed identity as a type statement: *vector* $\mathrm{d}\mathbf{r}$ equals *unit vector* $\mathbf{T}$ times *scalar* $\mathrm{d}s$. The unit vector carries the direction; the scalar carries the length.

2.3 The scalar area element $\mathrm{d}S$

Type: scalar. $\mathrm{d}S$ is an infinitesimal *area*, a non-negative number.

For a curve we measured the length of one tangent vector. For a surface we measure the *area of the parallelogram* spanned by two tangent vectors. Parametrize the surface by two variables, $\mathbf{r}(u, v)$. Moving u a little traces the tangent vector $\mathbf{r}_u\, \mathrm{d}u$; moving v a little traces $\mathbf{r}_v\, \mathrm{d}v$. The area of the parallelogram they span is the magnitude of their cross product:

$$\mathrm{d}S = \big|\mathbf{r}_u \times \mathbf{r}_v\big|\, \mathrm{d}u\, \mathrm{d}v.$$

Here $\mathbf{r}_u = \partial\mathbf{r}/\partial u$ and $\mathbf{r}_v = \partial\mathbf{r}/\partial v$ are vectors, their cross product is a vector, and its magnitude is a scalar. Compare directly:

$$\underbrace{\mathrm{d}s = |\mathbf{r}'(t)|\, \mathrm{d}t}_{\text{length of a tangent vector}} \qquad \longleftrightarrow \qquad \underbrace{\mathrm{d}S = |\mathbf{r}_u \times \mathbf{r}_v|\, \mathrm{d}u\, \mathrm{d}v}_{\text{area of a tangent parallelogram}} \quad .$$

One dimension uses the magnitude of a derivative; two dimensions use the magnitude of a cross product of partial derivatives. Same idea, one rung up.

2.4 The vector area element dS

Type: vector. $\mathrm{d}\mathbf{S}$ is an infinitesimal area carrying a *direction*: it points perpendicular to the surface, with magnitude equal to $\mathrm{d}S$.

This is the surface analogue of $\mathrm{d}\mathbf{r} = \mathbf{T}\, \mathrm{d}s$. The cross product $\mathbf{r}_u \times \mathbf{r}_v$ is already perpendicular to the surface, so we keep it whole instead of taking its magnitude:

$$\mathrm{d}\mathbf{S} = (\mathbf{r}_u \times \mathbf{r}_v)\, \mathrm{d}u\, \mathrm{d}v = \mathbf{n}\, \mathrm{d}S, \qquad \mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|} \ \text{ (unit normal).}$$

Again read it as types: *vector* $\mathrm{d}\mathbf{S}$ equals *unit vector* $\mathbf{n}$ times *scalar* $\mathrm{d}S$. The normal $\mathbf{n}$ is what makes "flux through a surface" meaningful: it records which way you are passing through.

Geometry	**Scalar element**	**Vector element**	**Link**
curve	$\mathrm{d}s = \lvert\mathbf{r}'\rvert\, \mathrm{d}t$	$\mathrm{d}\mathbf{r} = \mathbf{r}'\, \mathrm{d}t$	$\mathrm{d}\mathbf{r} = \mathbf{T}\, \mathrm{d}s$
surface	$\mathrm{d}S = \lvert\mathbf{r}_u \times \mathbf{r}_v\rvert\, \mathrm{d}u\, \mathrm{d}v$	$\mathrm{d}\mathbf{S} = (\mathbf{r}_u \times \mathbf{r}_v)\, \mathrm{d}u\, \mathrm{d}v$	$\mathrm{d}\mathbf{S} = \mathbf{n}\, \mathrm{d}S$

2.5 The volume element $\mathrm{d}V$

Type: scalar. $\mathrm{d}V$ is an infinitesimal *volume*. In Cartesian coordinates $\mathrm{d}V = \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z$. Under a change of variables $\mathbf{r}(u, v, w)$ it picks up the absolute value of the Jacobian determinant,

$$\mathrm{d}V = \big|\det[\, \mathbf{r}_u\ \mathbf{r}_v\ \mathbf{r}_w\,]\big|\, \mathrm{d}u\, \mathrm{d}v\, \mathrm{d}w,$$

which is the three-dimensional version of the parallelogram-area idea: the determinant of the three tangent vectors is the (signed) volume of the parallelepiped they span.

3 The three operators as typed functions

The symbols ∇, $\nabla\cdot$, and $\nabla\times$ look like three unrelated notations. They are better understood as one object, ∇, combined with the three products a vector can participate in. Write ∇ as a "vector of partial-derivative operators",

$$\nabla = \Big(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\Big).$$

This is a type pun, but a productive one: ∇ behaves like a vector whose components happen to be operators.

Operator	Product pattern	Signature	Result type
$\operatorname{grad} f = \nabla f$	scalar $\times$ vector	$(\mathbb{R}^3 \to \mathbb{R}) \to (\mathbb{R}^3 \to \mathbb{R}^3)$	vector field
$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$	vector $\cdot$ vector	$(\mathbb{R}^3 \to \mathbb{R}^3) \to (\mathbb{R}^3 \to \mathbb{R})$	scalar field
$\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F}$	vector $\times$ vector	$(\mathbb{R}^3 \to \mathbb{R}^3) \to (\mathbb{R}^3 \to \mathbb{R}^3)$	vector field

The result type is forced by the product, exactly as in ordinary vector algebra: a scalar times a vector is a vector; a dot product of two vectors is a scalar; a cross product of two vectors is a vector. If you remember the output types of the three products, you remember the output types of grad, div, and curl for free.

3.1 Component formulas

Let f be a scalar field and $\mathbf{F} = (P, Q, R)$ a vector field, with $P, Q, R : \mathbb{R}^3 \to \mathbb{R}$.

Gradient (scalar field $\to$ vector field). Each component is a partial derivative; the field of steepest ascent:

$$\nabla f = \Big(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\Big).$$

Divergence (vector field $\to$ scalar field). The dot product $\nabla \cdot \mathbf{F}$ sums the matching partials; it measures net outflow per unit volume:

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

Curl (vector field $\to$ vector field). The cross product $\nabla \times \mathbf{F}$ is the formal determinant

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \Big(R_y - Q_z,\quad P_z - R_x,\quad Q_x - P_y\Big),$$

where subscripts denote partial derivatives. It measures local rotation.

A note on dimension. Divergence and gradient make sense in any $\mathbb{R}^n$. Curl as written is special to $\mathbb{R}^3$ (the cross product lives there). In $\mathbb{R}^2$ one uses the *scalar curl* $Q_x - P_y$, which is the $\mathbf{k}$-component above; this is the quantity that appears in Green's theorem.

4 The integrals

We now assemble field $\times$ element = integrand, and read off the type at each step. In every case the final move is a parametrization that turns the integral into an ordinary $\int(\cdots)\,\mathrm{d}t$ or $\iint(\cdots)\,\mathrm{d}u\,\mathrm{d}v$.

4.1 Scalar line integral: $\int_C f\,\mathrm{d}s$

Signature 4.1. $\displaystyle\int_C f\,\mathrm{d}s$: (scalar field f, curve C) $\to$ scalar.

Integrand type. f is a scalar field, evaluated along the curve it returns a scalar; $\mathrm{d}s$ is a scalar. Scalar times scalar is scalar. Integrating scalars gives a scalar (e.g. the mass of a wire with density f).

Unpacking to one variable. Substitute $\mathrm{d}s = |\mathbf{r}'(t)|\,\mathrm{d}t$ from Section 2.1 and evaluate f at the moving point $\mathbf{r}(t)$:

$$\int_C f\,\mathrm{d}s = \int_a^b f\big(\mathbf{r}(t)\big)\,|\mathbf{r}'(t)|\,\mathrm{d}t.$$

The right-hand side is an ordinary single-variable integral: integrand $f(\mathbf{r}(t))|\mathbf{r}'(t)|$ is a plain function of t.

4.2 Vector line integral (work): $\int_C \mathbf{F}\cdot \mathrm{d}\mathbf{r}$

Signature 4.2. $\displaystyle\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r}$: (vector field $\mathbf{F}$, oriented curve C) $\to$ scalar.

Integrand type. $\mathbf{F}$ is a vector field (returns a vector); $\mathrm{d}\mathbf{r}$ is a vector; the dot product collapses the two vectors to a scalar. So although both inputs are vectors, the integrand is a *scalar*, and the integral is a number (the work done by $\mathbf{F}$ along C).

Unpacking to one variable. Substitute $\mathrm{d}\mathbf{r} = \mathbf{r}'(t)\,\mathrm{d}t$:

$$\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r} = \int_a^b \mathbf{F}\big(\mathbf{r}(t)\big)\cdot\mathbf{r}'(t)\,\mathrm{d}t.$$

Inside, $\mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)$ is a dot product of two concrete vectors, hence a scalar function of t, and you integrate it normally.

Connection to the scalar line integral. Using $\mathrm{d}\mathbf{r} = \mathbf{T}\,\mathrm{d}s$,

$$\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r} = \int_C (\mathbf{F}\cdot\mathbf{T})\,\mathrm{d}s.$$

So a work integral is just the scalar line integral of the *tangential component* $\mathbf{F}\cdot\mathbf{T}$ (a scalar field) of the force. The vector integral and the scalar integral are the same machine fed differently.

In components, with $\mathbf{F} = (P, Q, R)$ and $\mathrm{d}\mathbf{r} = (\mathrm{d}x, \mathrm{d}y, \mathrm{d}z)$, the same integral is often written

$$\int_C P\,\mathrm{d}x + Q\,\mathrm{d}y + R\,\mathrm{d}z,$$

which is literally the dot product $\mathbf{F}\cdot\mathrm{d}\mathbf{r}$ written out term by term.

4.3 Scalar surface integral: $\iint_S f\,\mathrm{d}S$

Signature 4.3. $\displaystyle\iint_S f\,\mathrm{d}S$: (scalar field f, surface S) $\to$ scalar.

Integrand type. f returns a scalar; $\mathrm{d}S$ is a scalar; product is scalar (e.g. mass of a sheet with surface density f).

Unpacking to two variables. Substitute $\mathrm{d}S = |\mathbf{r}_u \times \mathbf{r}_v|\,\mathrm{d}u\,\mathrm{d}v$:

$$\iint_S f\,\mathrm{d}S = \iint_D f\big(\mathbf{r}(u,v)\big)\,|\mathbf{r}_u\times\mathbf{r}_v|\,\mathrm{d}u\,\mathrm{d}v.$$

The right-hand side is an ordinary double integral over the flat parameter region $D \subset \mathbb{R}^2$, which you evaluate as two nested single-variable integrals.

4.4 Flux integral: $\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S}$

Signature 4.4. $\displaystyle\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S}$: (vector field $\mathbf{F}$, oriented surface S) $\to$ scalar.

Integrand type. $\mathbf{F}$ returns a vector; $\mathrm{d}\mathbf{S}$ is a vector; the dot product makes the integrand a scalar (the flux: how much of $\mathbf{F}$ passes through S).

Unpacking to two variables. Substitute $\mathrm{d}\mathbf{S} = (\mathbf{r}_u \times \mathbf{r}_v)\,\mathrm{d}u\,\mathrm{d}v$:

$$\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S} = \iint_D \mathbf{F}\big(\mathbf{r}(u,v)\big) \cdot (\mathbf{r}_u \times \mathbf{r}_v)\,\mathrm{d}u\,\mathrm{d}v.$$

The integrand is a dot product of concrete vectors, a scalar function of (u, v), and the result is an ordinary double integral. Using $\mathrm{d}\mathbf{S} = \mathbf{n}\,\mathrm{d}S$ this is also $\iint_S (\mathbf{F} \cdot \mathbf{n})\,\mathrm{d}S$: the scalar surface integral of the normal component $\mathbf{F} \cdot \mathbf{n}$.

4.5 Volume integral: $\iiint_V f\,\mathrm{d}V$

Signature 4.5. $\displaystyle\iiint_V f\,\mathrm{d}V$: (scalar field f, solid V) $\to$ scalar.

Both factors are scalars; the result is a scalar. In Cartesian coordinates it is just three nested single-variable integrals, $\iiint_V f\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$. There is no vector version: a region in $\mathbb{R}^3$ has no leftover direction to carry, so $\mathrm{d}V$ is intrinsically scalar.

4.6 Summary of integrand types

Integral	Field type	Element type	Integrand
$\int_C f\,\mathrm{d}s$	scalar	scalar (length)	scalar
$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$	vector	vector (tangent)	scalar (via $\cdot$)
$\iint_S f\,\mathrm{d}S$	scalar	scalar (area)	scalar
$\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S}$	vector	vector (normal)	scalar (via $\cdot$)
$\iiint_V f\,\mathrm{d}V$	scalar	scalar (volume)	scalar

Notice the pattern: every one of these integrals returns a *scalar*. The vector inputs always get reduced to a scalar integrand by a dot product before integration. When you see "$\cdot$" next to a differential, read it as "two vectors enter, one scalar leaves."

5 The integral theorems: one pattern

The four big theorems all say the same structural thing: *integrating a derivative over a region equals integrating the original field over the boundary of that region.* Each step also drops the dimension of integration by one. This is the multidimensional Fundamental Theorem of Calculus.

The one-dimensional FTC is the template:

$$\int_a^b f'(x)\,\mathrm{d}x = f(b) - f(a).$$

Left side: a derivative integrated over the interval $[a, b]$. Right side: the original function evaluated on the boundary $\{a, b\}$, which is a 0-dimensional set (two points, with signs). Every theorem below is this sentence with the region upgraded.

5.1 Gradient theorem (FTC for line integrals)

$$\int_C \nabla f \cdot \mathrm{d}\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a}), \qquad C:\ \mathbf{a} \to \mathbf{b}.$$

Types. Left: ∇f is a vector field, dotted with $\mathrm{d}\mathbf{r}$ to give a scalar line integral over the curve C (a 1-dimensional region). Right: the scalar field f evaluated on the boundary $\partial C = \{\mathbf{a}, \mathbf{b}\}$ (two points, 0-dimensional), with signs. The derivative ∇ inside the 1-dimensional integral becomes the bare field f on the 0-dimensional boundary.

5.2 Green's theorem (in the plane)

$$\oint_{\partial D} \big(P\,\mathrm{d}x + Q\,\mathrm{d}y\big) = \iint_D \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)\,\mathrm{d}A.$$

Types. Right: the scalar curl $Q_x - P_y$ (a scalar field) integrated over the plane region D (2-dimensional). Left: the vector field (P, Q) integrated along the boundary curve ∂D (1-dimensional) as a work integral. The derivative combination is inside on the area; the plain field is on the boundary curve.

5.3 Stokes' theorem

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathrm{d}\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot \mathrm{d}\mathbf{r}.$$

Types. Left: $\nabla \times \mathbf{F}$ is a vector field (the curl), turned into a scalar flux integrand by $\cdot\,\mathrm{d}\mathbf{S}$, integrated over the surface S (2-dimensional). Right: $\mathbf{F}$ itself integrated along the boundary curve ∂S (1-dimensional) as work. Stokes is Green's theorem lifted off the plane onto a curved surface; Green's is the special case where S is a flat region and $\mathbf{F}$ has no z-component.

5.4 Divergence theorem

$$\iiint_V (\nabla \cdot \mathbf{F})\,\mathrm{d}V = \iint_{\partial V} \mathbf{F} \cdot \mathrm{d}\mathbf{S}.$$

Types. Left: $\nabla \cdot \mathbf{F}$ is a scalar field (the divergence), integrated over the solid V (3-dimensional). Right: $\mathbf{F}$ as a flux through the boundary surface ∂V (2-dimensional). The derivative is inside on the volume; the plain field flows through the boundary.

5.5 The unifying table

Reading down the "inside" column you always find a derivative of the field; reading down the "boundary" column you always find the plain field. Reading across, the dimension of the integral drops by exactly one.

Theorem	dim	Inside region (derivative)	dim	On boundary (field)
FTC	1	$\int_a^b f'\,\mathrm{d}x$	0	$f(b) - f(a)$
Gradient thm	1	$\int_C \nabla f \cdot \mathrm{d}\mathbf{r}$	0	$f(\mathbf{b}) - f(\mathbf{a})$
Green / Stokes	2	$\iint_S (\nabla \times \mathbf{F}) \cdot \mathrm{d}\mathbf{S}$	1	$\oint_{\partial S} \mathbf{F} \cdot \mathrm{d}\mathbf{r}$
Divergence	3	$\iiint_V (\nabla \cdot \mathbf{F})\,\mathrm{d}V$	2	$\iint_{\partial V} \mathbf{F} \cdot \mathrm{d}\mathbf{S}$

In the language of differential forms all four lines are literally one equation, $\int_\Omega \mathrm{d}\omega = \int_{\partial\Omega} \omega$, where d is a single operator (the exterior derivative) that specializes to grad, curl, and div depending on the type of ω. You do not need that machinery to use the theorems, but it explains why the four statements rhyme: they are one statement, type-cast to dimensions 1, 2, and 3.

6 Reading checklist

When you meet an unfamiliar vector-calculus expression, recover its types in this order.

1. **Identify the region and its dimension.** Curve (C, 1D), surface (S, 2D), or solid (V, 3D)? The number of integral signs usually matches.
2. **Identify the differential element and its type.** $\mathrm{d}s, \mathrm{d}S, \mathrm{d}V$ are scalars (lengths, areas, volumes). $\mathrm{d}\mathbf{r}, \mathrm{d}\mathbf{S}$ are vectors (they carry a direction: tangent and normal respectively).
3. **Identify the field and its type.** Lowercase f: scalar field. Boldface $\mathbf{F}$: vector field. A "$\cdot$" or "$\times$" nearby signals a vector field is involved.
4. **Combine and check the integrand is a scalar.** Scalar $\times$ scalar, or vector $\cdot$ vector. If you have a vector next to a vector element with a dot, the dot makes it scalar. The integral itself is then a number.
5. **Cast to ordinary calculus.** Parametrize: substitute $\mathrm{d}s = |\mathbf{r}'|\,\mathrm{d}t$, $\mathrm{d}\mathbf{r} = \mathbf{r}'\,\mathrm{d}t$, $\mathrm{d}S = |\mathbf{r}_u \times \mathbf{r}_v|\,\mathrm{d}u\,\mathrm{d}v$, or $\mathrm{d}\mathbf{S} = (\mathbf{r}_u \times \mathbf{r}_v)\,\mathrm{d}u\,\mathrm{d}v$, evaluate the field at the moving point, and integrate the resulting plain function of t (or u, v).

The single sentence to carry away: *a parametrization is a type cast from a geometric integral to the one-variable integral you already know how to do.*